

NEST: Locally Certified Search for Structural-Entropy Hierarchies

Dingli Su¹, Yicheng Pan^{1(✉)}, and Angsheng Li^{1,2}

¹ State Key Laboratory of Complex & Critical Software Environment, Beihang University, Beijing, China

{sudingli,yichengp,angsheng}@buaa.edu.cn

² Zhongguancun Laboratory, Beijing, China

Abstract. Structural entropy evaluates a graph hierarchy through the description length of an encoding tree. HCSE and BBM [12] construct such trees but do not determine whether the result admits a local entropy-reducing topology change. We introduce NNI-certified Entropy Search over Trees (NEST), a multi-start optimizer based on rooted nearest-neighbor interchange (NNI). For weighted graphs, we derive the exact NNI entropy change and prove that an improving rotation can be inaccessible to a monotone merge or compression. The formula supports best-improvement descent and a checkable one-NNI local-optimality certificate. Because a better tree may require a temporary increase, NEST also tests bounded two-move paths but commits only improving endpoints. We also formulate an $O(3^n)$ subset dynamic program, using binary sufficiency [14, 12], to compute exact global optima for audit graphs. Across 500 graphs from five 64-vertex hierarchical regimes, NEST lowers entropy by 0.4239 ± 0.0086 bits relative to the better of HCSE and BBM and wins all 500 paired comparisons. In sealed, candidate-budget-matched audits, 32 coalescent starts reach the exact global optimum on 249/250, 99/100, and 22/25 instances at $n = 12, 14, 16$, respectively (370/375 overall), whereas 32-call HCSE and label-free BBM reach 2/375 and 1/375. These finite audits measure observed optimality on the declared suites; they do not establish an approximation ratio.

Keywords: Structural entropy · hierarchical clustering · tree editing · nearest-neighbor interchange · graph algorithms

1 Introduction

Many networks are organized at more than one scale: vertices form local groups, related groups form larger modules, and the desired output is therefore a tree rather than one flat partition. Structural entropy supplies an information-theoretic objective for such an encoding tree [7]. Recent HCSE and BBM methods construct low-entropy hierarchies through recursive partitioning and balanced merging [12]. Their construction rules leave a complementary question unanswered: *after construction, can a local topology change further reduce structural entropy?*

A hierarchy constructor can make an early merge that later steps never revisit, even when rotating a nearby internal edge would lower structural entropy. We therefore seek a topology operator whose effect can be computed exactly, whose termination certifies the inspected neighborhood, and whose output can be checked against a true optimum on exactly solvable instances.

Nearest-neighbor interchange (NNI) is the elementary rotation of a rooted binary tree. It changes one internal edge while preserving every leaf. NNI is standard in evolutionary-tree search and has been studied for other hierarchical-clustering objectives, but its structural-entropy change has not been used as an exact graph-hierarchy optimizer. We show that, for $((A, B), C) \rightarrow (A, (B, C))$, all objective terms cancel except two cross-module edge weights and three module volumes. This identity yields monotone descent and a one-NNI local-optimality certificate.

Our method, NNI-certified Entropy Search over Trees (NEST), refines several inexpensive initial trees with the exact operator and returns the lowest independently verified endpoint. A bounded two-move beam can cross a small intermediate barrier, but accepts a pair only when its endpoint lowers the full objective. A randomized-restart variant, NEST-R, increases basin coverage without inspecting labels or the exact optimum. Using prior binary sufficiency [14, 12], we specialize an exact dynamic program over vertex subsets to compute global optima for small-graph audits. The resulting pipeline connects an exact local identity, a certified search procedure, and an independent global benchmark.

Our contributions are:

- an exact weighted-graph NNI delta, a proof of separation from monotone single merge/compress steps, and a checkable one-NNI local certificate;
- NEST, which combines multi-start construction, exact NNI refinement, and bounded two-move escape without weakening the output guarantee;
- an SE-specific $O(3^n)$ arbitrary-subset trellis optimizer for small graphs, built on prior binary sufficiency, and an exact audit of NEST’s optimality gap; and
- a paired evaluation against SE agglomeration, Louvain-2L, Paris, and the official HCSE and BBM implementations, with entropy, hierarchy recovery, runtime, memory, operator audits, and component ablations.

NEST wins all 500 paired comparisons with HCSE and BBM, and 32 coalescent starts reach the exact optimum on 370/375 audited graphs.

2 Preliminaries

Weighted graph and encoding tree. Let $G = (V, E, w)$ be an undirected graph with nonnegative edge weights. For $S \subseteq V$, let $V_S = \text{vol}(S) = \sum_{u \in S} d_u$, and let g_S be the total weight of edges with exactly one endpoint in S . The total graph volume is $\text{vol}(V) = \sum_u d_u = 2 \sum_{e \in E} w_e$. An encoding tree T has one leaf per vertex. Each internal node a represents the union S_a of its descendant leaves; a^- denotes its parent.

The structural entropy of G under T is

$$H^T(G) = - \sum_{a \neq r} \frac{g_{S_a}}{\text{vol}(V)} \log_2 \frac{V_{S_a}}{V_{S_a^-}}. \quad (1)$$

Our sole optimization objective is Eq. (1); lower is better. No ground-truth label is used to construct or select a tree. For a zero-volume module, its summand is defined as zero; such a module has no incident positive-weight edge and can be attached after optimizing the positive-volume part of the tree.

Constructors. We distinguish a *constructor*, which produces an initial tree, from a *refiner*, which searches neighboring topologies. Our candidate pool contains (i) a binary SE-agglomerative dendrogram, (ii) a recursive SE partition tree, and (iii) a Paris dendrogram when the optional implementation is available [2]. HCSE and BBM are distinct external constructors and are evaluated from their official implementation [12].

Rooted NNI. An NNI move acts on a binary parent-child edge. Here A , B , and C denote three disjoint subtrees, and $P = A \cup B \cup C$ denotes their shared local parent module: the set of all leaves contained in those three subtrees. Writing the local topology as $((A, B), C)$, the two nontrivial alternatives are $(A, (B, C))$ and $(B, (A, C))$. The leaf set, graph, and all modules outside P remain unchanged. A rooted binary tree with n labeled leaves has finitely many topologies, namely $(2n - 3)!!$, so any strictly decreasing local search terminates.

3 NNI-Certified Entropy Search

NEST builds several initial hierarchies, improves each through NNI rotations, and returns the tree with the lowest structural entropy. The local identity of Section 3.1 scores every candidate rotation exactly, while a full recomputation of Eq. (1) checks every accepted tree. An exact dynamic program supplies the global optimum for the small-graph audits.

3.1 A local structural-entropy identity

For disjoint sets X, Y , write $W(X, Y) = \sum_{u \in X, v \in Y} w_{uv}$. Consider the rotation shown in Fig. 1. The disjoint subtrees A, B, C fill their local parent $P = A \cup B \cup C$; the rotation replaces its child module $X = A \cup B$ with $Y = B \cup C$ while leaving the rest of the hierarchy unchanged.

Theorem 1 (Exact weighted NNI delta). *For an eligible rotation with $V_X, V_Y, V_P > 0$, the change in tree structural entropy is*

$$\Delta H^T = \frac{2}{\text{vol}(V)} \left[W(A, B) \log_2 \frac{V_P}{V_X} + W(B, C) \log_2 \frac{V_Y}{V_P} \right]. \quad (2)$$

The symmetric alternative follows by exchanging A and B .

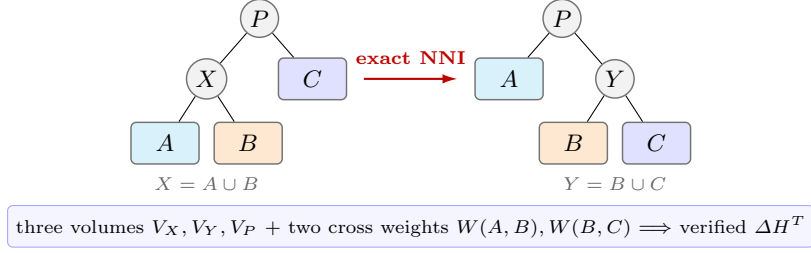


Fig. 1. A rooted NNI replaces only $X = A \cup B$ by $Y = B \cup C$. The highlighted local statistics determine the exact full-objective change; all leaves and the surrounding tree remain fixed.

Proof. Deferred to Appendix A.

Proposition 1 (Atomic advantage over merge-compress). *Let T_c contract the edge from P to $X = A \cup B$, exposing A, B, C as siblings, and let T' then merge B, C under $Y = B \cup C$. For these HCSE compress and merge primitives,*

$$H^{T_c}(G) - H^T(G) = \frac{2W(A, B)}{\text{vol}(V)} \log_2 \frac{V_P}{V_X} \geq 0, \quad (3)$$

$$H^{T'}(G) - H^{T_c}(G) = -\frac{2W(B, C)}{\text{vol}(V)} \log_2 \frac{V_P}{V_Y} \leq 0. \quad (4)$$

Their sum is Eq. (2). Thus NNI lowers entropy when the merge gain in Eq. (4) exceeds the compression loss in Eq. (3), although neither desired reparenting nor an improving compression is available as one primitive.

Proof. Deferred to Appendix A.

This proves strict separation from monotone single-primitive search, not global dominance over a complete HCSE round, which may cross an intermediate increase. We evaluate Eq. (2) from cached sets and sparse adjacency lists, then verify every selected move by fully recomputing Eq. (1).

3.2 One-step certificate and compound escape

Proposition 2 (Monotonicity and local certificate). *One-step refinement never increases $H^T(G)$. If it stops before applying M moves, every rooted NNI on an eligible binary parent-child edge of the returned tree has $\Delta H^T \geq -\varepsilon$: the tree is ε -locally optimal. With $\varepsilon = 0$ in exact arithmetic, no improving rooted NNI remains.*

Proof. Deferred to Appendix A. We use $\varepsilon = 10^{-10}$ bits, a floating-point tolerance, $M = 100$, and the exact identity of Theorem 1.

Strict one-step descent can stop before a better tree separated by a shallow barrier. Algorithm 1 therefore keeps a short list—a beam—of the b most promising first moves whose increase is at most β , then evaluates a second NNI from each intermediate tree. It never commits the intermediate tree alone.

Algorithm 1: Exact NNI refinement of one candidate tree

Input: Graph G , tree T , tolerance ε , move budget M , barrier β , beam b , compound rounds R

```

1 repeat
2   evaluate Eq. (2) for every legal NNI of  $T$ ;
3   apply the most negative move and verify the full objective;
4 until every legal NNI has  $\Delta H \geq -\varepsilon$ , or  $M$  moves were applied;
5 for at most  $R$  compound rounds do
6   retain the  $b$  first moves with smallest  $\Delta H$  and  $\Delta H \leq \beta$ ;
7   expand every legal second NNI from each retained intermediate tree;
8   if no two-move endpoint has entropy below  $H^T(G) - \varepsilon$  then
9     break;
10  commit the best endpoint; rerun one-step descent;
11 return  $T$ 

```

Proposition 3 (Safe compound search). *The compound stage never worsens its input. It can reach an improving endpoint even when every one-step neighbor is non-improving.*

Proof. Deferred to Appendix A.

A strict local trap. A one-NNI certificate does not imply global optimality. On a fixed eight-vertex weighted instance, one-step descent terminates at a tree with entropy 1.970038 bits because every legal NNI neighbor has a nonnegative objective change. Reaching a better tree requires first crossing a 0.013970-bit barrier. A second rotation, followed by ordinary one-step descent, then reaches 1.920593 bits, reducing the original objective by 0.049445 bits. Thus the bounded two-move stage can leave a certified one-step local minimum, while the endpoint check ensures that NEST commits only a tree with lower entropy. The complete instance and move sequence are included with the artifact and protected by a regression test.

3.3 Fast NNI-certified entropy search

NEST applies Algorithm 1 independently to SE-agglomerative, recursive-SE, and Paris starts and returns the candidate with minimum verified entropy. The initializer is recorded for every output, so gains can be attributed to candidate selection, one-step refinement, and compound escape.

Binary and multiway trees. SE agglomeration, Paris, and coalescent starts are binary; recursive SE, HCSE, BBM, and Louvain-2L may contain multiway nodes. We do not binarize them: NNI acts only on binary parent-child edges, so for a multiway tree the certificate of Proposition 2 covers only those edges. Binarizing could only help, because grouping two children A, B of a node P under a new node S changes the entropy by $-2W(A, B) \log_2(V_P/V_S)/\text{vol}(V) \leq 0$ (Proposition 4).

In the 500-graph benchmark every tree returned by NEST was binary (333 from SE agglomeration, 167 from Paris), and a direct check confirmed the ε -certificate on all 500/500 of them.

Complexity. With n leaves there are at most $2(n-2)$ oriented NNI alternatives per sweep. Each delta needs two cross weights, obtained in $O(m)$ time by scanning the adjacency lists of the smaller module; cached leaf sets take $O(nh)$ space, where h is tree height. A committed move is verified in $O(m \log h + n)$ time. One-step descent therefore costs $O(Mn(m+h))$, and each compound round costs $O(bn(m \log h + n))$, because every retained first move is rescored and followed by a full second sweep. The total is $O((M+Rb)n(m \log h + n))$ time and $O(nh+m)$ space per initializer. The budgets did not bind in the benchmark: the returned trees needed at most 41 descent moves ($M = 100$) and 5 compound rounds ($R = 8$).

Parameters. The largest barrier crossed by any accepted two-move escape was 0.008 bits, well below $\beta = 0.05$; on the 50-graph ablation subset, $\beta \in \{0.01, 0.2\}$ and $R \in \{2, 32\}$ left the result unchanged. The beam width is the only binding setting: $b = 4$ raises mean entropy by 0.0014 bits (21/50 graphs worse), while $b = 64$ lowers it by 0.0055 bits (31/50 graphs better), while the compound cost grows linearly in b . We keep $b = 16$ as a compromise.

Randomized basin coverage. NEST-R adds q coalescent starts, each built by repeatedly merging two uniformly selected current components, then applies the same NNI routine. The distribution is not uniform over labeled topologies. Selection uses only rescored entropy; increasing q cannot worsen the result, and its global-hit rate is measured against exact optima below.

3.4 Exact global audit on small graphs

To measure how often certified endpoints coincide with global optima, we use a separate exact solver. SuperTAD proves binary sufficiency for general graphs, and HCSE also notes it [14, 12]; we restate the result to make the audit self-contained.

Proposition 4 (Binary sufficiency). *Every encoding tree has a binary refinement whose structural entropy is no greater. Consequently, a global minimum exists among rooted binary trees.*

Proof. Included for completeness in Appendix A.

For a nonempty vertex subset S , let $F(S)$ be the minimum contribution of a binary subtree rooted at S , excluding the edge from S to its parent. Then

$$F(\{v\}) = 0, \quad F(S) = \min_{A \dot{\cup} B = S} \left\{ F(A) + F(B) - \frac{g_A}{\text{vol}(V)} \log_2 \frac{V_A}{V_S} - \frac{g_B}{\text{vol}(V)} \log_2 \frac{V_B}{V_S} \right\}. \quad (5)$$

Enumerating each unordered split once gives $O(3^n)$ time and $O(2^n)$ memory, an SE-specific subset dynamic program, equivalently a trellis whose states are vertex subsets [9]. By Proposition 4, $F(V)$ is the global minimum over all encoding trees, not only over the binary subclass. The exact audits apply this routine at $n = 12, 14, 16$; the deployable NEST algorithm does not invoke it.

4 Experiments

4.1 Questions and protocol

The experiments test the consequences of the exact local identity and the search procedure derived from it. We ask: (Q1) how often do established constructors leave an improving NNI; (Q2) does compound search add improvements beyond a one-step certificate; (Q3) does lower entropy preserve or improve recovery of a planted hierarchy; and (Q4) how close is NEST to the global optimum on exactly solvable graphs?

The sealed main protocol uses 100 graph seeds per regime. Every method sees the identical weighted graph, every tree is rescored by the same independent implementation of Eq. (1), and we report paired means with 95% t -intervals. The compound search uses beam width 16, barrier $\beta = 0.05$ bits, and at most eight accepted compound rounds. No label is used by NEST for model selection. BBM is additionally shown with the planted number of fine clusters, which is an oracle advantage and is marked as such.

Data. Our controlled suite contains five two-level hierarchical stochastic block models: clean, noisy, imbalanced, weighted, and weak-hierarchy. Each balanced graph has 64 vertices, eight fine blocks, and four coarse blocks; the imbalanced regime has the same total size with block sizes from 4 to 12. The generator manifest, any 0.01-weight connectivity edges, and every seed are stored with the results. This family follows the hierarchical stochastic block model used to study recovery beyond worst-case inputs [3].

Methods. We compare SE agglomeration, a two-level Louvain tree [1], Paris [2], official HCSE, official BBM with oracle k [12], and NEST, whose pool (SE agglomeration, recursive SE, Paris) excludes HCSE, BBM, and Louvain-2L; these remain external comparators. The operator audit separately applies NNI to every constructor’s returned tree.

Metrics. The primary metric is $H^T(G)$. Dendrogram purity is computed for the planted fine and coarse partitions: for every same-class vertex pair, it measures the class fraction under the pair’s predicted LCA, then averages over pairs. For the exact suite we report the relative gap $100(H_{\text{NEST}}^T - H^*)/H^*$ and the *optimal-hit rate*: the fraction of graphs whose objective is within $\max(10^{-9}, 10^{-8}|H^*|)$ of the exact optimum.

Table 1. Mean tree structural entropy H^T (bits) on 100 paired graphs per HSBM regime. The $\pm$ value is a 95% t -interval over graph seeds; lower is better, and bold marks the lowest column mean. $\text{BBM}^\dagger$ receives the planted fine-cluster count; the other methods do not use labels.

Method	Clean	Noisy	Imbalanced	Weighted	Weak hierarchy
SE-agglomerative	2.899 ± 0.021	3.723 ± 0.017	3.162 ± 0.023	2.445 ± 0.016	3.461 ± 0.022
Louvain-2L	4.109 ± 0.022	4.603 ± 0.013	4.241 ± 0.019	3.657 ± 0.032	4.432 ± 0.018
Paris	2.869 ± 0.022	3.709 ± 0.017	3.129 ± 0.023	2.349 ± 0.015	3.438 ± 0.023
HCSE	3.228 ± 0.031	4.485 ± 0.068	3.598 ± 0.036	2.741 ± 0.029	4.032 ± 0.051
$\text{BBM}^\dagger$	3.335 ± 0.021	4.118 ± 0.020	3.573 ± 0.025	3.007 ± 0.020	3.924 ± 0.022
NEST (ours)	2.827 ± 0.021	3.683 ± 0.017	3.086 ± 0.023	2.340 ± 0.015	3.414 ± 0.021

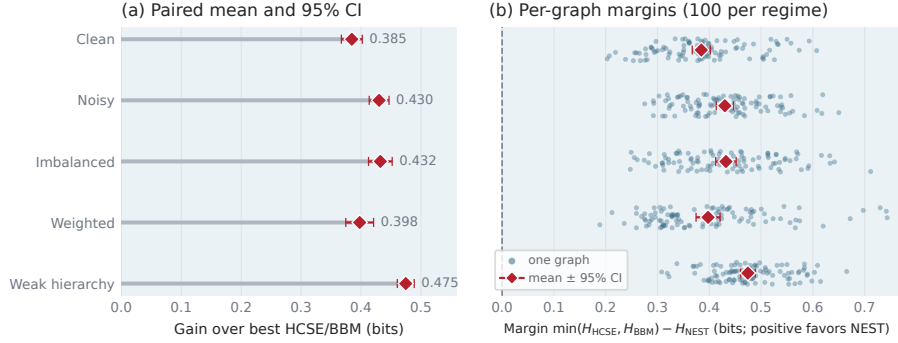


Fig. 2. Paired advantage over external structural-entropy methods. For graph g , the margin is $M_g = \min(H_{\text{HCSE},g}^T, H_{\text{BBM},g}^T) - H_{\text{NEST},g}^T$, so positive values favor NEST. (a) Regime mean and paired 95% confidence interval. (b) Each blue dot is one of 100 sealed graphs per regime; vertical jitter only separates overlapping dots. Red diamonds are regime means and horizontal whiskers are their 95% confidence intervals. All 500 margins are positive.

4.2 Frozen benchmark results

NEST *returns the lowest-entropy tree on every one of the 500 graphs*. Table 1 compares the tree returned by each standalone baseline with the best verified endpoint of NEST’s three-initializer pool; the NEST row does not post-process HCSE or BBM outputs. NEST has the lowest mean H^T in all five regimes. Relative to HCSE and BBM, its paired reductions are 0.5468 ± 0.0218 and 0.5213 ± 0.0101 bits. Figure 2 plots, for each graph g , the margin $M_g = \min(H_{\text{HCSE},g}^T, H_{\text{BBM},g}^T) - H_{\text{NEST},g}^T$ against the stronger baseline; all 500 margins are positive. BBM’s internal clustering is unseeded: rerunning 15 of these graphs changed its entropy by at most 0.10 bits, below the smallest margin of 0.19 bits. NEST, HCSE, Paris, and both SE constructors reproduced bit-for-bit.

Lower entropy coincides with better planted-hierarchy recovery on average, but not on every graph (Q3). Fine-level dendrogram purity is 0.546 for NEST, versus

Table 2. NNI refinement audit over 500 graphs. Drop is mean reduction relative to raw H^T ; improved is the fraction of graphs with a strict reduction. Two-move values are additional to one-NNI descent, and time excludes initial tree construction.

Constructor	One-NNI descent		Two-move escape		Refinement time
	Entropy drop	Graphs improved	Extra entropy drop	Graphs further improved	(s)
SE-agglomerative	1.97%	100%	0.14%	62%	0.642
Paris	0.07%	74%	0.06%	41%	0.388
HCSE	0.00%	3%	0.00%	4%	0.108
BBM [†]	14.02%	100%	0.46%	93%	1.962

0.483 for HCSE and 0.486 for oracle- k BBM; the paired gains are 0.062 ± 0.005 and 0.060 ± 0.007 . Coarse-level gains are 0.081 ± 0.007 and 0.110 ± 0.010 . NEST’s fine purity is nonetheless lower than HCSE’s on 61/500 graphs, so entropy is not a proxy for recovery on an individual instance.

4.3 The operator audit

Constructor quality and NNI-local optimality are different properties (Q1, Q2). Table 2 applies the same two-stage refinement to each constructor’s output: one-NNI descent commits the best decreasing move until none remains, and the two-move escape then accepts only a lower-entropy endpoint of a bounded pair. One-step NNI improves every SE-agglomerative and oracle-BBM tree, and after that certificate the two-move escape still improves 62% and 93% of them. On 71/500 BBM trees descent reached the $M = 100$ move budget, so the BBM drops are lower bounds. Paris trees are improved in 74% and then 41% of graphs. HCSE trees are nearly NNI-locally optimal, with rates of 3% and 4%, yet their mean entropy exceeds that of Paris in every regime of Table 1.

4.4 Components and exact optimality

Every component of NEST contributes. On a separate frozen 50-graph subset (full ablation table in Table 8), starting from SE agglomeration, the candidate pool improves 34 graphs, one-NNI descent then improves 45, and the two-move escape adds gains on 30. The main benchmark’s verifier checks hashes, schema, paired seeds, unique keys, and monotonicity of committed NNI endpoints across its 3,500 method records and 500 generator manifests.

On exactly solvable graphs, 32 coalescent starts reach the global optimum in 370/375 cases (Q4). Disjoint calibration fixed $q = 32$; we then sealed 250 graphs at $n = 12$, 100 at $n = 14$, and 25 at $n = 16$ across the same five regimes. Every method receives exactly 32 candidates: coalescent starts for NEST; eight seeded HCSE calls at each target height 2–5; and label-free BBM calls cycling through $k = 2, \dots, 8$ (four or five per k). The deterministic NEST pool is excluded from this budget. HCSE is nearly deterministic, returning at most 3 distinct entropies

Table 3. Sealed exact-optimum audit on independently generated HSBMs with n vertices and 32 candidates per method: coalescent starts for NEST, target heights for HCSE, and label-free cluster counts for BBM. Exact columns report optimum hits over the listed graphs. The NEST interval is an exact 95% Clopper–Pearson interval for its hit rate. Its relative gap is $100(H_{\text{NEST}}^T - H^*)/H^*$, reported as mean/worst; zero is optimal. Candidate selection does not use H^* .

n	Graphs	Exact optima found			NEST diagnostics	
		NEST	HCSE	BBM	95% hit-rate interval	Mean / worst relative gap
12	250	249/250	2/250	1/250	[97.79, 99.99]%	0.00032% / 0.07990%
14	100	99/100	0/100	0/100	[94.55, 99.97]%	0.00121% / 0.12060%
16	25	22/25	0/25	0/25	[68.78, 97.45]%	0.03990% / 0.44934%

Table 4. Tree structural entropy H^T (bits; lower is better) on four real networks: Zachary karate club, Florentine families, Les Misérables, and Davis Southern women. Bold marks the lowest value per network. BBM[†] uses ground-truth k for Karate and Louvain’s k otherwise.

Method	Karate	Florent.	Les Mis.	Davis
SE-agglomerative	2.698	1.923	3.078	3.055
Louvain-2L	3.405	2.452	3.923	3.813
Paris	2.597	1.923	2.920	2.981
HCSE	2.929	2.172	3.930	3.206
BBM [†]	3.330	2.254	3.199	3.387
NEST (ours)	2.586	1.919	2.916	2.962

per graph, whereas BBM returns 3–23. Selection uses only structural entropy, and the exact dynamic program is consulted afterward. Table 3 reports 2/375 exact hits for HCSE and 1/375 for label-free BBM, with exact binomial intervals for NEST at each size. The interval widens to [68.78, 97.45]% at $n = 16$, where only 25 graphs were audited. Every block has status zero, matching SHA-256 seals, complete seed coverage, and independently recomputed summaries; a fresh rerun of 15 of these graphs reproduced every H^* and every NEST outcome.

4.5 Real networks

NEST *also attains the lowest entropy on 4/4 real networks* (Table 4). These networks lack a unique ground-truth hierarchy, so this experiment measures objective minimization only. Paris is the closest competitor, 0.004–0.019 bits behind.

5 Related Work

Structural entropy and hierarchy construction. Li and Pan introduced structural information as an encoding-tree measure of network organization [7]. HCSE and

BBM develop hierarchy construction and balancing operations for this objective [12]. SuperTAD proves binary sufficiency and uses it in contiguity-constrained dynamic programs [14]; HCSE also notes the property [12]. Our exact audit instead optimizes arbitrary vertex subsets. The structural-entropy survey organizes the broader theory, optimization, and application literature [13]. NEST asks whether a returned tree is locally optimal under an exact topology rotation and uses the answer as both a diagnostic and an optimizer.

Graph hierarchical clustering. Paris supplies an inexpensive agglomerative graph hierarchy related to modularity [2]. Louvain optimizes modularity [1]; we score its final partition as a two-level tree. Dasgupta’s cost and its revenue dual have inspired hierarchical algorithms with guarantees [4, 11]. Because these objectives differ from structural entropy, optimizing Eq. (1) requires its own local identity.

Exact inference and heuristic accuracy. Cluster trellises support exact hierarchical inference [9]. Our routine is an SE specialization: the prior binary-sufficiency property justifies the subset recurrence, while our arbitrary-subset formulation differs from SuperTAD’s contiguity-constrained dynamic programs.

Local search in tree space. NNI is a classical neighborhood for evolutionary, or phylogenetic, trees; its reconfiguration distance is itself nontrivial [8]. Recent work studies NNI local search for hierarchical-clustering costs [5]. PERCH uses purity-enhancing rotations during online tree construction, while GRINCH combines rotations with longer-range grafts for general linkage functions [6, 10]. Our formula instead targets weighted graph structural entropy and exposes the two cross-module weights controlling an NNI rotation, and evaluates the complete objective on arbitrary weighted graphs.

6 Limitations and Conclusion

NNI is a local operator, and its certificate is not one of global optimality. The two-move escape explores a bounded neighborhood, and multiway regions must be resolved by a constructor before binary rotations can act on them. Lower entropy raised planted purity on average but not on every graph, and no coalescent restart recovered the strict planted tree in the exact and sampled basin audits at $n = 5$ –8 and $n = 12$. The exact solver is exponential, so the near-optimality measurements cover only the declared 12–16-vertex suites.

We introduced NEST, which uses an exact weighted NNI identity for monotone refinement, a local-optimality certificate, and bounded escape from shallow traps. It had the lowest entropy on all 500 paired graphs and reached the exact optimum on 370/375 sealed instances at $n = 12, 14, 16$, versus 2/375 for HCSE and 1/375 for label-free BBM; this does not imply global optimality beyond the audited suites.

Code and data availability. The implementation, sealed per-graph records, verifiers, and table generators are available at <https://github.com/SuuTTT/nest-tamc-2026>.

Acknowledgments. This work was supported by the National Natural Science Foundation of China (62576021).

References

1. Blondel, V.D., Guillaume, J.L., Lambiotte, R., Lefebvre, E.: Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment* **2008**(10), P10008 (2008). <https://doi.org/10.1088/1742-5468/2008/10/P10008>
2. Bonald, T., Charpentier, B., Galland, A., Hollocou, A.: Hierarchical graph clustering using node pair sampling. In: *Advances in Neural Information Processing Systems*. pp. 8214–8223 (2018)
3. Cohen-Addad, V., Kanade, V., Mallmann-Trenn, F.: Hierarchical clustering beyond the worst-case. In: *Advances in Neural Information Processing Systems*. vol. 30 (2017)
4. Dasgupta, S.: A cost function for similarity-based hierarchical clustering. In: *Proceedings of STOC*. pp. 118–127 (2016). <https://doi.org/10.1145/2897518.2897527>
5. Jowhari, H.: Hierarchical clustering via local search. *arXiv preprint arXiv:2405.15983* (2024)
6. Kobren, A., Monath, N., Krishnamurthy, A., McCallum, A.: A hierarchical algorithm for extreme clustering. In: *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. pp. 255–264 (2017). <https://doi.org/10.1145/3097983.3098079>
7. Li, A., Pan, Y.: Structural information and dynamical complexity of networks. *IEEE Transactions on Information Theory* **62**(6), 3290–3339 (2016)
8. Li, M., Tromp, J., Zhang, L.: On the nearest neighbour interchange distance between evolutionary trees. *Journal of Theoretical Biology* **182**(4), 463–467 (1996)
9. Macaluso, S., Greenberg, C., Monath, N., Lee, J.A., Flaherty, P., Cranmer, K., McGregor, A., McCallum, A.: Cluster trellis: Data structures and algorithms for exact inference in hierarchical clustering. In: *Proceedings of AISTATS*. vol. 130, pp. 2467–2475. PMLR (2021)
10. Monath, N., Kobren, A., Krishnamurthy, A., Glass, M.R., McCallum, A.: Scalable hierarchical clustering with tree grafting. In: *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. pp. 1438–1448 (2019). <https://doi.org/10.1145/3292500.3330929>
11. Moseley, B., Wang, J.R.: Approximation bounds for hierarchical clustering: Average linkage, bisecting k-means, and local search. In: *Advances in Neural Information Processing Systems*. vol. 30, pp. 3094–3103 (2017)
12. Pan, Y., Fan, B., Long, P., Zheng, F.: An information-theoretic perspective of hierarchical clustering on graphs. In: *Proceedings of UAI*. vol. 286, pp. 3322–3345. PMLR (2025)
13. Su, D., Peng, H., Pan, Y., Li, A.: A survey of structural entropy: Theory, methods, and applications. In: *Proceedings of IJCAI*. pp. 10660–10668 (2025)
14. Zhang, Y.W., Wang, M.B., Li, S.C.: SuperTAD: Robust detection of hierarchical topologically associated domains with optimized structural information. *Genome Biology* **22**(1), 45 (2021). <https://doi.org/10.1186/s13059-020-02234-6>

A Proof Details and Additional Basin Analysis

This appendix supplies proofs omitted from the 12-page main paper, the component ablation, and additional finite-instance evidence; none of its claims is needed to interpret the main experimental tables.

Proof of Theorem 1. First suppose the three child modules have positive volume; the zero-volume case follows by deleting modules with no incident positive-weight edge and taking the zero-contribution limit. Only the terms for A, B, C, X , and Y can change. Omitting the common factor $1/\text{vol}(V)$, their old and new contributions are

$$\begin{aligned} L_{\text{old}} &= g_A \log_2 \frac{V_X}{V_A} + g_B \log_2 \frac{V_X}{V_B} + g_X \log_2 \frac{V_P}{V_X} + g_C \log_2 \frac{V_P}{V_C}, \\ L_{\text{new}} &= g_A \log_2 \frac{V_P}{V_A} + g_B \log_2 \frac{V_Y}{V_B} + g_C \log_2 \frac{V_Y}{V_C} + g_Y \log_2 \frac{V_P}{V_Y}. \end{aligned}$$

Using $g_X = g_A + g_B - 2W(A, B)$ and $g_Y = g_B + g_C - 2W(B, C)$, the coefficients of g_A, g_B, g_C telescope to zero. The remaining terms are

$$2W(A, B) \log_2(V_P/V_X) + 2W(B, C) \log_2(V_Y/V_P).$$

Restoring $1/\text{vol}(V)$ proves Eq. (2). External boundary weights and $W(A, C)$ do not enter the delta.

Proof of Proposition 1. Delete X 's codeword and substitute $g_X = g_A + g_B - 2W(A, B)$ in Eq. (1); this gives Eq. (3). The symmetric calculation for inserting Y gives Eq. (4).

Proof of Proposition 2. Only moves with $\Delta H^T < -\varepsilon$ are committed, and their endpoint is independently verified, so the objective never increases. The loop exits only when the best enumerated move has $\Delta H^T \geq -\varepsilon$ or after M moves; in the first case every legal move on an eligible binary parent-child edge has $\Delta H^T \geq -\varepsilon$.

Proof of Proposition 3. A pair is committed only after its final tree is fully recomputed and found to have lower entropy than the tree preceding the pair. The first move is not committed independently, so its sign cannot weaken the output guarantee. A two-edge path may have a positive first delta and a negative net delta; the regression witness in Sec. 3.2 instantiates this case.

Proof of Proposition 4 (included for completeness). At a node P , choose two children with disjoint leaf sets A, B , insert their union $S = A \cup B$ below P , and make A, B children of S . The objective change is

$$-\frac{2W(A, B)}{\text{vol}(V)} \log_2 \frac{V_P}{V_S} \leq 0.$$

Repeated insertion converts every multiway node to binary without increasing the objective.

Table 5. Exact restart-basin probabilities for one fixed noisy HSBM at each size. The All topologies column gives $(2n - 3)!!$, the number of unordered rooted binary labeled starting trees. Standard pool exact reports whether NEST’s deterministic initializer pool reaches the global optimum H^* . p_U is the fraction of uniformly weighted starts refined to H^* ; p_C uses NEST’s pairwise-coalescent restart distribution. Strict recovery requires every planted fine and coarse block to appear as a clade.

n	All topologies	Standard pool exact?	Optimum basin		Strict-recovery basin
			Uniform $p_U(H^*)$	Coalescent $p_C(H^*)$	Coalescent $p_C(\text{strict})$
5	105	yes	100.000%	100.000%	0.000%
6	945	yes	100.000%	100.000%	0.000%
7	10,395	yes	100.000%	100.000%	0.000%
8	135,135	no	24.848%	25.237%	0.000%

A.1 Exact and sampled basin probabilities

The finite hit rates in the main paper do not supply a per-start probability. For a fixed graph G , deterministic refinement map F , optimum H^* , and start measure μ , that probability is

$$p_G = \sum_T \mu(T) \mathbf{1}\{H^{F(T)}(G) = H^*(G)\}.$$

It depends on the graph, start distribution, tie-breaking, and refinement configuration. The exact subset DP supplies H^* but not p_G .

For $n \leq 8$ we enumerate all $(2n - 3)!!$ unordered rooted binary labeled topologies. Uniform topology mass is not NEST’s restart distribution. Under the pairwise coalescent, a topology with child subtrees of a, b leaves has compatible-history count

$$h(T) = \binom{a+b-2}{a-1} h(T_A) h(T_B), \quad \mu(T) = \frac{h(T)}{\prod_{k=2}^n \binom{k}{2}}.$$

The history counts sum exactly to the denominator. Table 5 shows that the selected hard $n = 8$ graph has exact coalescent basin mass $p_G = 0.252365$; because the standard pool misses it, 32 independent starts give $1 - (1 - p_G)^{32} = 0.999909$, not certainty.

A.2 Detailed 12-vertex method audit

Table 6 retains the full method and runtime breakdown for the largest exact-audit stratum. The main paper reports the cross-size comparison.

At $n = 12$, exact enumeration would require 13,749,310,575 topologies. Table 7 therefore uses direct Bernoulli sampling rather than the earlier first-hit diagnostic. The estimates vary substantially by graph: 8.07%–46.43% per start. On noisy-11 this implies only 93.23% for 32 random starts (a conservative 95% lower value of

Table 6. Detailed budget-matched audit on 250 independently generated 12-vertex HSBMs. Candidates/graph is the number proposed before structural-entropy-only selection; H^* is revealed afterward. The successful graphs column counts completed runs, whereas exact-optimum rates use all 250 graphs. Relative gap is $100(H^T - H^*)/H^*$ (mean $\pm$ 95% CI over successful graphs), and zero is best. Time includes all candidates for one graph.

Method	Candidates per graph	Successful graphs	Exact optima count (rate)	Mean relative gap $\pm$ 95% CI (%)	Mean time per graph (s)
NEST (coalescent)	32	250/250	249 (99.6%)	0.000320 ± 0.000629	0.171
HCSE	32	250/250	2 (0.8%)	10.902 ± 0.971	0.024
BBM (oracle k)	32	248/250	1 (0.4%)	15.054 ± 1.13	0.402
BBM (label-free)	32	250/250	1 (0.4%)	10.237 ± 0.593	0.299
SE agglomerative	1	250/250	44 (17.6%)	1.764 ± 0.268	< 0.001

Table 7. Restart success on eight hard $n = 12$ graphs, using 10,000 independent coalescent starts per graph. Graph IDs are regime-seed. $\hat{p}(H^*)$ is the fraction of starts refined to the exact optimum; intervals are exact 95% Clopper-Pearson intervals. $q_{32} = 1 - (1 - \hat{p})^{32}$ estimates the chance that at least one of 32 independent starts succeeds; lower q_{32} substitutes the lower confidence limit for $\hat{p}$. Strict planted recovery occurred 0/10,000 times on every graph (per-graph 95% upper bound 0.0369%).

Graph	Successful starts $\hat{p}(H^*)$	95% interval for $\hat{p}$	Predicted 32-start success q_{32}	Conservative lower q_{32}
noisy-0	17.19%	[16.46, 17.94]%	99.761%	99.683%
noisy-8	14.61%	[13.92, 15.32]%	99.362%	99.175%
noisy-9	20.56%	[19.77, 21.37]%	99.937%	99.913%
imbal.-0	46.43%	[45.45, 47.41]%	100.000%	100.000%
weak-9	30.61%	[29.71, 31.52]%	99.999%	99.999%
noisy-11	8.07%	[7.54, 8.62]%	93.229%	91.872%
noisy-17	17.75%	[17.01, 18.51]%	99.808%	99.743%
weak-17	17.86%	[17.11, 18.63]%	99.816%	99.754%

91.87%), explaining why the observed finite-suite success must not be read as a probabilistic guarantee.

Objective success is deliberately separated from strict planted recovery. No one of the 80,000 hard-case endpoints contained every planted fine and coarse block as a clade; the exact $n = 8$ enumeration likewise gives zero mass. This strict event is stronger than dendrogram purity and exposes objective-recovery misalignment on these small realized HSBMs. Thus the restart evidence supports optimization of H^T , not a claim that every entropy optimum equals the generator’s planted hierarchy.

Component ablation. Table 8 gives the cumulative ablation summarized in Section 4.

Refinement runtime. Figure 3 complements Table 2 with the quality-time trade-off of each initializer.

Table 8. Cumulative NEST ablation on 50 paired HSBMs (10 per regime). Each + row adds one component to the preceding row. The candidate pool selects the lowest-entropy initial tree among SE agglomeration, recursive SE, and Paris. $\Delta H^T = H_{\text{previous}}^T - H_{\text{row}}^T$, so positive values are improvements; the final column counts graphs with a strict decrease.

Variant	Mean H^T (bits)	Mean ΔH^T (bits)	Graphs improved
SE agglomeration	3.1435	—	—
+ candidate pool	3.1041	0.0394	34/50
+ one-NNI descent	3.0794	0.0247	45/50
+ two-move escape	3.0759	0.0035	30/50

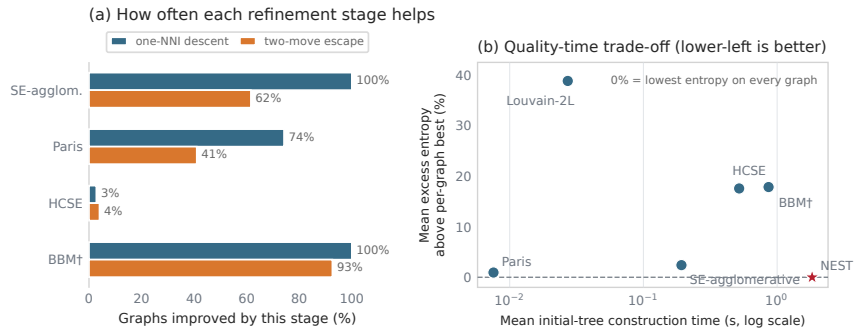


Fig. 3. NNI refinement over 500 graphs. (a) Blue is the fraction of initial trees improved by repeated best-decreasing NNI moves. After no single move helps, the two-move escape may cross one uphill intermediate tree; orange is the fraction whose final endpoint improves further. (b) Each point is an initializer. Excess entropy is $100(H^T - H_{\text{best}}^T)/H_{\text{best}}^T$, where H_{best}^T is the lowest value on the same graph; thus zero is best. The horizontal axis is mean construction time in log seconds, so lower-left is better. Times include Python allocation tracing.

A graph-only lower bound. For an additional graph-only certificate, an edge-LCA expansion gives

$$H^T(G) = \sum_{uv \in E} \frac{w_{uv}}{\text{vol}(V)} \log_2 \frac{V_{\text{lca}(u,v)}^2}{d_u d_v} \geq \sum_{uv \in E} \frac{w_{uv}}{\text{vol}(V)} \log_2 \frac{(d_u + d_v)^2}{d_u d_v}. \quad (6)$$

The inequality follows from $V_{\text{lca}(u,v)} \geq d_u + d_v$. It is a universal lower bound, but the exact dynamic program gives the sharper certificates reported below.